

Exploratory Data Analysis

1. Dataset Overview

Three datasets were loaded:

- **Books_Cleaned.csv** – Contains metadata about books.
- **Users_Cleaned.csv** – Includes user demographic data.
- **Ratings_Cleaned.csv** – Contains user ratings for books.

Each dataset was inspected for:

- Column names and data types using `df.info()`
- Missing values using `df.isnull().sum()`
- Summary statistics via `df.describe()`

2. Exploratory Data Analysis Summary

i. Dataset Inspection:

- No critical missing values noted.
- All columns were typed and named appropriately.

ii. User Behaviour:

- Average ratings per user were calculated.
- Users were segmented based on the number of books rated:
 - Users rating >10, >20, >50 books identified.
- This can help filter active users for collaborative filtering later.

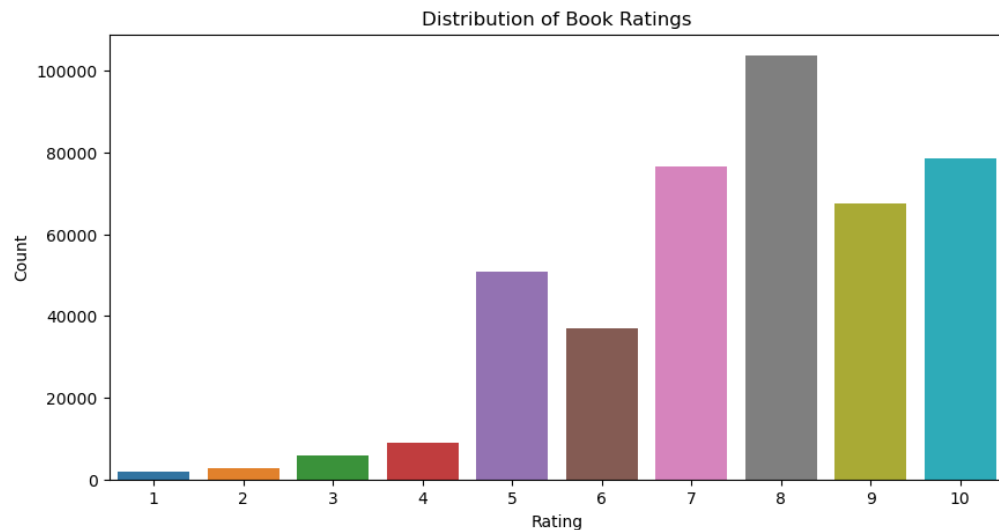
iii. Book Behaviour:

- Average rating per book was analyzed.
- Identified:
 - Books with the highest average ratings
 - Books with lowest average ratings (potentially controversial or niche)

- Also useful for recommending trending or polarizing books.

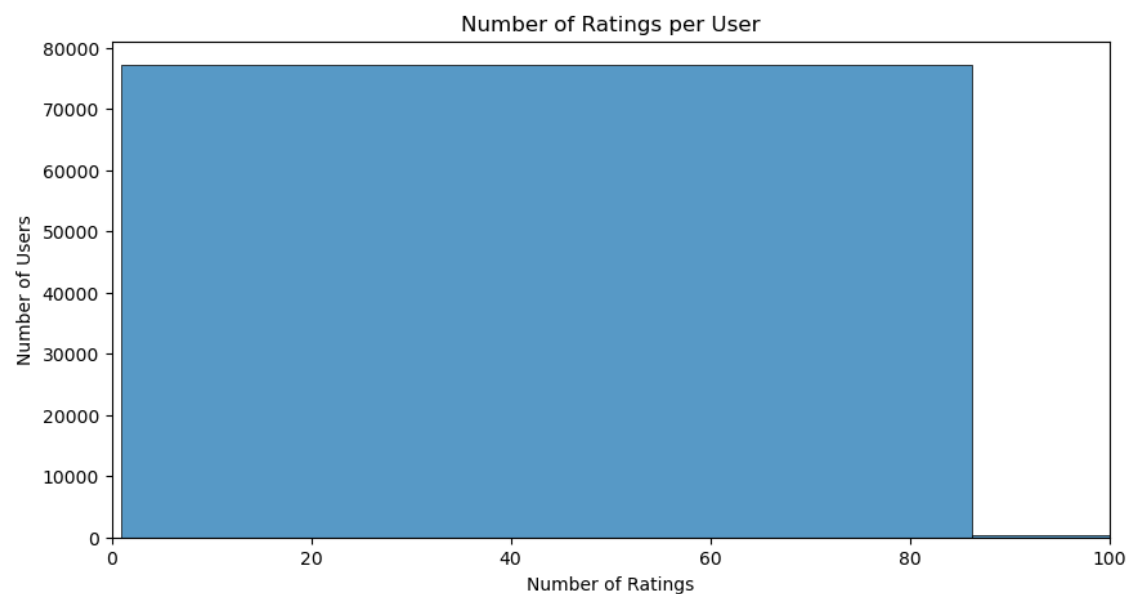
iv. Rating Distribution:

- Used `seaborn.countplot()` to visualize the distribution of ratings.
- Ratings are typically on a 1–10 scale.



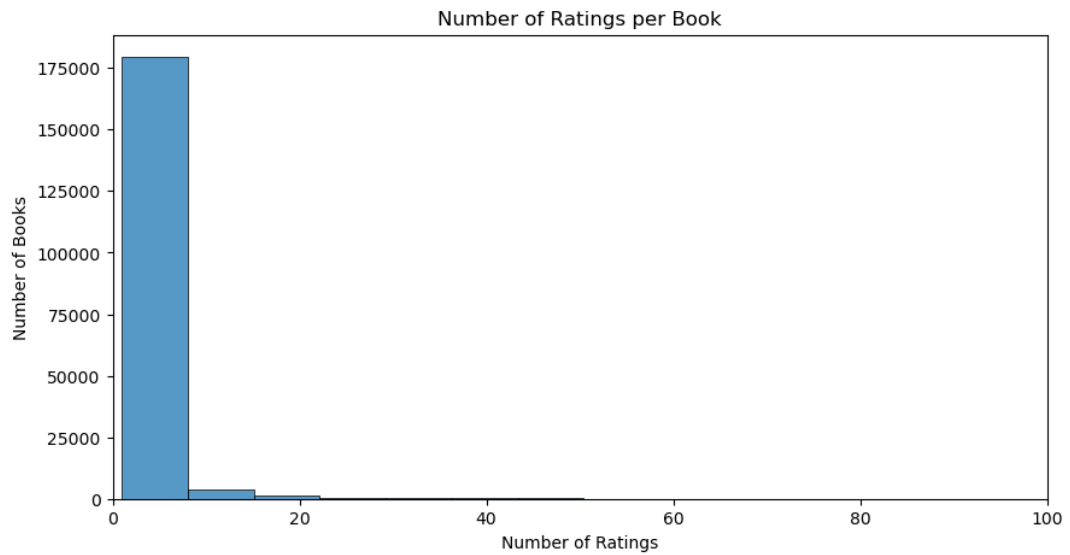
- Insights:
 - Most users gave mid-to-high ratings.
 - Lower ratings were less frequent, indicating a potential user bias toward favourable reviews.

v. Ratings per User:



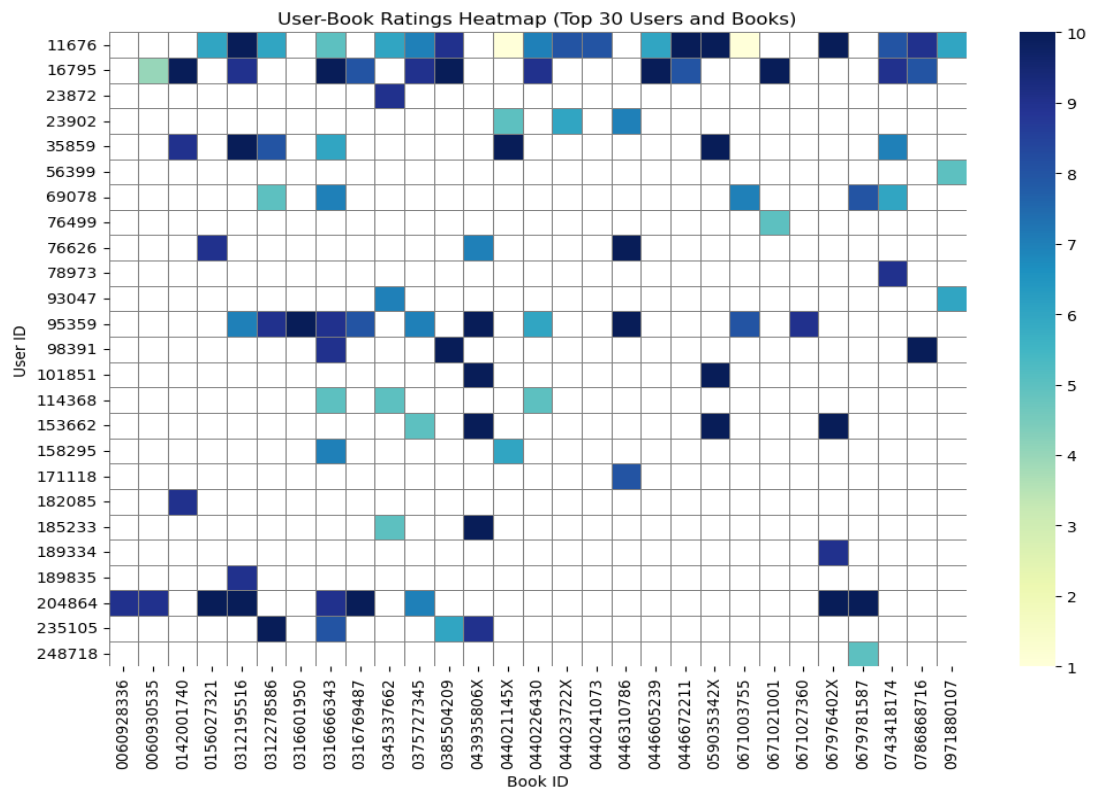
- Most users rate only a few books.
- A few power users rate a large number of books.

vi. Ratings per Book:



- Many books have very few ratings.
- A small number of books are very popular and heavily rated.

vii. user-book interactions (sparse matrix view)



viii. Top Users and Books:

Top 10 Most Active Users:

```
user_id
11676      8524
98391      5802
153662     1969
189835     1906
23902      1395
76499      1036
171118     1035
235105     1023
16795       968
248718      948
dtype: int64
```

Top 10 Most rated books:

```
book_id
0316666343    707
0971880107    581
0385504209    487
0312195516    383
0679781587    333
0060928336    320
059035342X    313
0142001740    307
0446672211    295
044023722X    281
dtype: int64
```

ix. Top 10 highest rated books with at least 50 ratings

Top 10 Highest Rated Books (with at least 50 ratings):

```
              mean  count
book_id
0345339738  9.402597     77
0439139597  9.262774    137
043936213X  9.207547     53
0345339711  9.120482     83
0439136369  9.082707    133
0064400557  9.073529     68
0439136350  9.035461    141
043935806X  9.033981    206
0590353403  8.983193    119
0156528207  8.980392     51
```

x. Data Sparsity:

- The user-book interaction matrix is sparse, with a sparsity level of 0.8932.
- High sparsity implies that most users have rated only a small subset of books, which can pose a challenge for collaborative filtering algorithms.